

Experiments on Guderley Mach reflection

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Abstract Experiments have been conducted in a large shock tube to examine the four-wave shock reflection pattern, now known as Guderley reflection (GR). The fourth wave, an expansion, is clearly identified, as is the supersonic patch behind the reflected wave. A shocklet terminating the supersonic patch behind the reflected wave is identified, which forms a second triple point further down the Mach stem. Evidence is presented showing the presence of more than one expansion wave and more than one shocklet, thus indicating the existence of more than one supersonic patch. In order to distinguish between cases with a single patch without the shocklet as originally proposed by Guderley and found in some computations, and the indications of a multi-patch geometry found here, and also in other computations, this latter case is designated Guderley Mach reflection (GMR). Multi-exposure images of the shock propagation superimposed on a single image frame enable estimates to be made of the strength of the major waves, and it is shown that the reflected wave is very weak.

Keywords Guderley reflection · Weak shocks · Supersonic patch · Von Neumann reflection

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1 Introduction

Prior to the past decade three distinct types of shock wave reflection were clearly identified for the reflection of weak to moderate strength shock waves off a wedge. These are known as regular reflection (RR), Mach reflection (MR) and von Neumann reflection (vNR). The equations which accurately describe the behaviour of the first two of these, known as the two- and three-shock theories, were developed by von Neumann [1] and are based on his oblique shock equations under the assumptions that the waves are infinitely thin and plane, and thus separate regions of uniform flow. However, for von Neumann reflection there is considerable disagreement between the experiments, which clearly show three waves, and the three-shock theory. This has come to be known as the von Neumann paradox. There have been a number of approaches in the past in an attempt to resolve the paradox, as briefly surveyed in [2].

Of particular relevance is a flow field proposed by Guderley [3,4] more than 50 years ago, in which he suggested a possible solution to the paradox for the case of a steady jet flow. This proposed the existence of a supersonic patch immediately behind the Mach stem (M) as shown in Fig. 1 with an expansion wave emanating from the triple point. The subsonic flow passing through the reflected wave (R) is converging and accelerating until it becomes sonic and a centered expansion results at the triple point (F). Thus the pattern at the triple point becomes a four-wave rather than a three-wave geometry. The reason that this hypothesis had been ignored for so long is that there had been no evidence experimentally, nor computationally, of the existence of the fourth wave up until very recently.

Some evidence that such a wave could exist has recently been shown in two groups of studies using high resolution computations with special treatment to resolve the flow at

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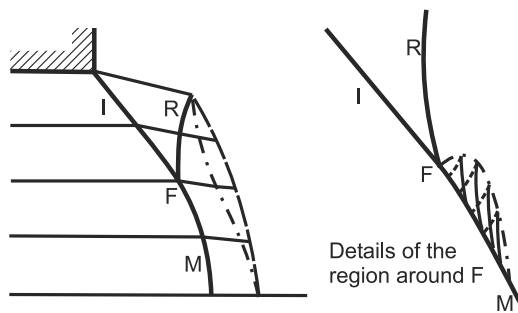


Fig. 1 The steady transonic jet flow pattern suggested by Guderley [3,4]

the triple point. A typical result from Vasil'ev and Kraiko [5] using the Euler equations is shown in Fig. 2a, showing a clearly defined supersonic patch and the expansion wave centered on the triple point. The second, which initially used the unsteady transonic small disturbance equations by Hunter and Brio [6] also showed the supersonic patch and the authors speculated that this could be terminated by a shock by analogy to that that occurs on a transonic aerofoil, and suggested that there could be a series of such patches. This was subsequently shown to be so by Tesdall and Hunter [7] using a new computational scheme with the result shown in Fig. 2b. The scheme in [5] showed a contact surface emanating from the triple point and a small region of supersonic flow below it, whereas [7] shows a succession of triple points along the Mach stem, and a very weak contact surface in [9]. All computations confirmed the extremely small size of these new features thereby indicating the reason that the four-wave system had not been identified experimentally. More recent extensions of these studies are given in [8] and [9]. The former identifies an interesting variation of Fig. 2a where the flow above the contact surface is as shown but that below is entirely subsonic. This variation was labeled ?R in [8] but is now commonly referred to as Vasilev reflection (VR), and is a subset of Guderley's original proposal. A theoretical model showing the boundaries between these two patterns is given in [8].

The above studies confirmed the basic Guderley four wave model although predicting somewhat different flow details, particularly on the possible existence of multiple patches. It is suggested in [8] that the multiple patch pattern is a transient phenomenon that is the result of flow unsteadiness. On the other hand both [5] and [9] solve the Euler equations in a pseudo-steady frame. The initial conditions for the two sets of simulations are, however, very different. Vasilev and Kraiko [5] use an incident shock Mach number of 1.47, specific heat ratio of 1.67, and Fig. 2a is for a wall angle of 12.5° . Tesdall et al. [9] uses 1.075, 1.4 and 15, respectively. It is thus possible that both these geometries are valid and that different patterns of reflection exist depending on the boundary conditions. For this reason it is appropriate to distinguish

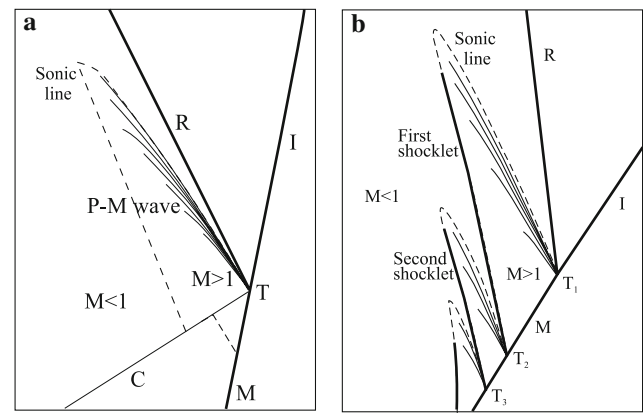


Fig. 2 The computed flow patterns of **a** Vasil'ev and Kraiko [5] and **b** Tesdall and Hunter [7]

between the two patterns of Fig. 2. This was suggested by a reviewer of this paper. The original proposal by Guderley matches the simulations in [5] in that there is no terminating shocklet to the supersonic patch, and thus this pattern should retain the designation of Guderley reflection (GR). The multiple patch system is distinctly different since the terminating shock to each patch results in a new triple point and a new four-wave pattern. Tesdall et al. [9] have used the term Guderley Mach reflection in their work so the designation (GMR) would seem appropriate. Clearly further study is needed to clarify this issue.

In view of the very small size of these patches relative to the length of the Mach stem experimental validation of the four wave pattern required the development of unique experimental facilities. Such confirmation was obtained by Skews and Ashworth [2] which clearly identified the fourth wave, as well as a shock terminating the supersonic patch as predicted in [7]. Some inconclusive indication of the existence of a second patch was found. The current work is an extension of that experimental investigation.

2 Apparatus

The apparatus comprises two main items: a unique shock tube specially constructed to generate a Mach reflection with a Mach stem length of the order of 1 m, together with its associated instrumentation, and a variety of optical systems to both capture high resolution individual images and a multi-exposure camera for obtaining quantitative data of the wave passage in space and time.

2.1 The shock tube facility

The rationale for the development of a special shock tube facility is given in [2]. The main difference from a

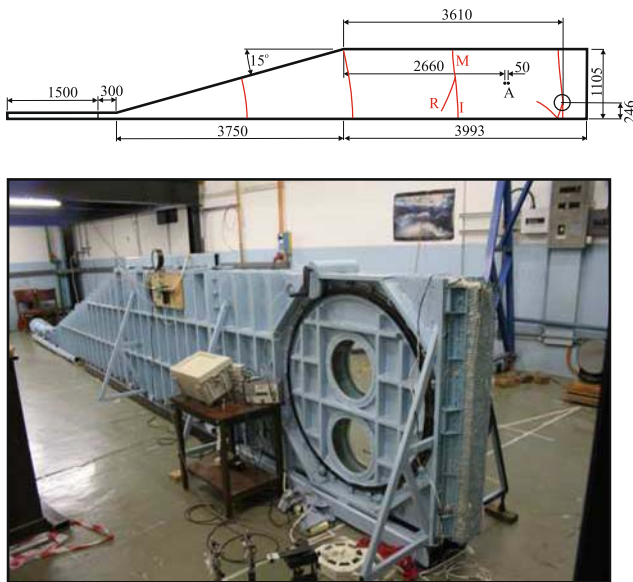


Fig. 3 The shock tube. Dimensions in mm

conventional tube is that the incident wave is cylindrical which at the window section where the photographs are taken has a radius of nearly 7.5 m. The basic sizes are given in the sketch of Fig. 3 together with a photograph of the facility. It is driven by a cylindrical cross-section driver, followed by a rectangular cross-section, tapered driven section along which the cylindrical wave propagates before striking the roof of the facility, which then results in the desired reflection pattern. Some typical wave profiles are shown in red on the sketch.

Two viewing ports are positioned on a large rotating insert in the sidewall of the tube in order to provide flexibility in capturing the wave system. The sketch and photograph show the position for the current series of tests. Tests were run at three nominal driver pressures of 1.8, 2.2 and 3.8 bar from which nominal wave transit Mach numbers were checked between two transducers positioned at A in the sketch. Higher pressures resulted in the reflection point occurring beyond the available window positions. Ambient pressure is 0.83 bar. Actual Mach numbers in the test window were determined from the multi-exposure imaging giving average triple point Mach numbers of 1.122, 1.129 and 1.167.

2.2 Photography

A conventional two-mirror Z-layout schlieren system was used for all imaging. Single-frame images were captured on a Fujifilm S3 digital camera having 12.34 million effective pixels. The multi-frame camera (Cooke Corporation) can take up to ten superimposed images with adjustable exposure times down to 1 μ s and variable interframe delay, at an effective pixel resolution, interlaced, of 768×493 . Initial single frame images were obtained using a Hammamatsu xenon flash lamp

with 15 W input and exposure time of 1.9 μ s using a 312 mm diameter schlieren beam. Once the position of the triple point was established to be reproducible, magnified images were obtained using a 100 mm diameter beam. To increase sensitivity the schlieren cutoff was made very small requiring a stronger light source, so this was changed to a 60 W input xenon flash lamp. Its longer exposure time of 2.9 μ s coupled with some motion blur due to the magnification resulted in the shock thickness in the image being increased but the reflection features that were captured were then large enough to almost fill the camera chip.

Substantial adjustment was needed to obtain better images compared to those presented in [2]. The overall illumination of the images varied considerably over the testing as adjustments were continually being made to the knife edges in an attempt to find the best balance of sensitivity and visibility. Although some variability was still experienced the vast majority of the few hundred images taken showed the primary supersonic patch and indications of the shocklet, when the knife edge is positioned parallel to the reflected wave. This wave is clearly very weak, as will be confirmed later, since it was not always detectable with other knife edge orientations. However, only tens of images gave a clear indication of further features, as will be discussed in Sect. 3.2. The reason for this has not been confirmed but may be due to vibrations in the system being transmitted through to the optical mounts due to diaphragm rupture and the rather substantial knife-edge cutoff.

No clear patches are detectable on the multi-frame images due to the much lower pixel resolution and the larger field of view, but the three main waves are sufficiently clear to obtain quantitative data of the wave velocities.

3 Supersonic patches

3.1 The primary patch

A typical image is given in Fig. 4. It is orientated in the conventional orientation for presenting shock reflection patterns with the undisturbed flow on the right and the incident shock at the top, and not in the inverted actual position shown in Fig. 3. The dark lines at the top and bottom of the image are cotton threads mounted across the windows positioned to give a scale to the image, in this case about 30 mm, and the corresponding Mach stem length would be about 800 mm. At this level of magnification the waves appear plane and there is a slight bend between the incident shock and the Mach stem at the triple point. The main issue, however, is that the patch is clearly visible with the expansion wave centered on the triple point and a weak terminating shock, similar to the primary patch predicted in [7]. There is some slight shading behind this shock suggesting a further expansion fan and

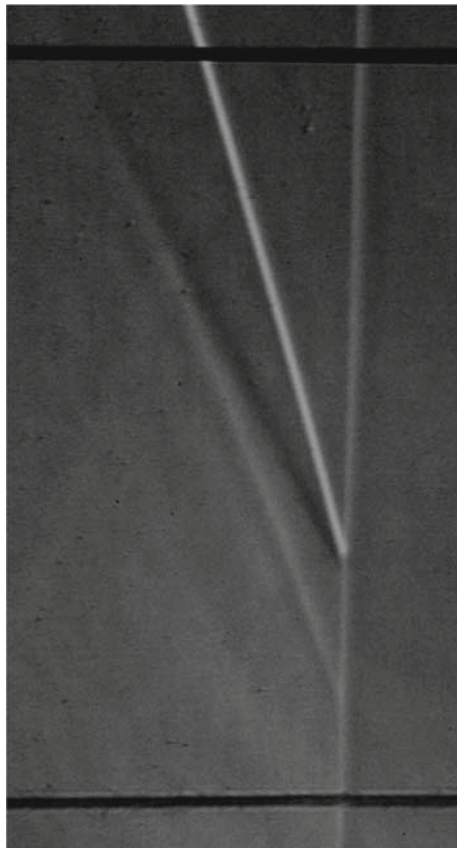


Fig. 4 Image of the primary supersonic patch (test M221900B, image 13-861) clearly showing the expansion wave centered on the triple point, and the terminating shock. Driver pressure of 2.2 bar

patch but this is clearly insufficient from which to draw any firm conclusions. No clear evidence of a slipstream is evident in the image, although the knife-edge orientation would mitigate against it being detected. Even with different knife-edge orientations it could not be positively identified.

Two further typical results using lower magnification are given in Fig. 5. This series of tests clearly show the existence of the expansion wave, confirming the existence of the fourth wave of a GR. However the extent of the patch and the terminal shock are not that evident and could easily be missed unless they were specifically being sought. This is somewhat different from Fig. 4 where these features are clearly defined.

3.2 Evidence of a second patch

With the shock tube and instrumentation resources available the chances of establishing whether more than one patch could exist was initially considered doubtful. Nevertheless, a fair number of images do show distinct features which indicate that it and even further disturbances behind the Mach stem do exist. A selection of these are given in Fig. 6 taken at three different optical magnifications. In the first image there

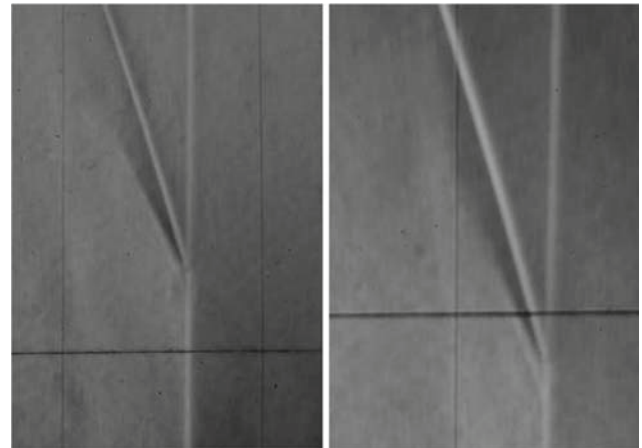


Fig. 5 Lower magnification images showing the clearly defined fourth wave of the Guderley reflecton as a black fan-shaped region immediately behind the reflected wave (test S222020A, image 19-1033, and test T221990A, image 20-1063, both taken with a driver pressure of 2.2 bar)

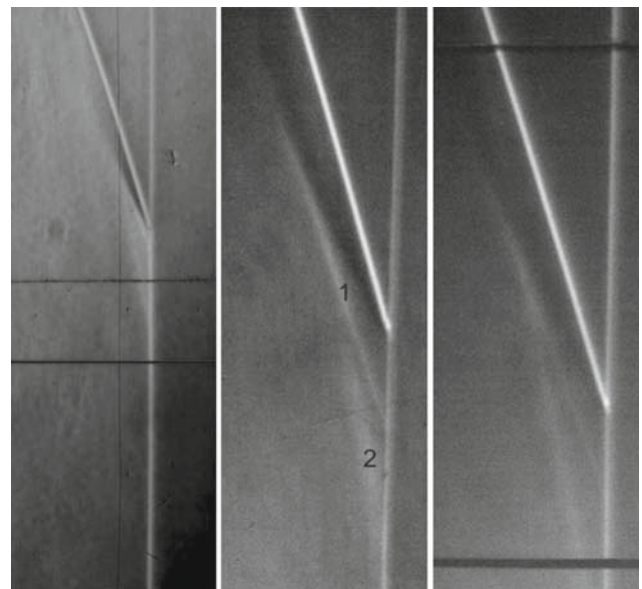


Fig. 6 Evidence of a second patch. The two patches are the lighter areas identified in the middle image (test N221970B image 14-0915; test P221980C image 16-0954; and test S222080A image 19-1036. All tests at 2.2 bar driver pressure)

is a distinct lighter coloured patch behind the shocklet whilst in the second and third this separates into a clearly distinguishable separate patch with a darker region separating it from the first patch.

In order to evaluate these differences in shading more clearly a further two images are modified by increasing contrast and then by plotting contours around a selected value of grayscale in the image. These images show much less of an indication of further features than in Fig. 6. This was done on purpose to make them more representative of the bulk of

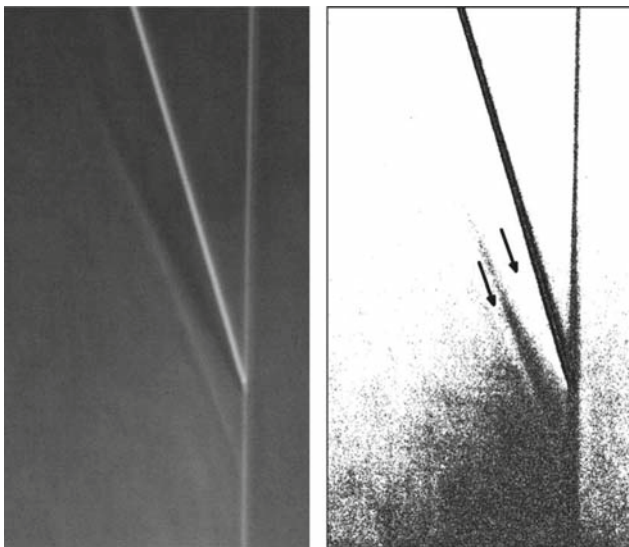


Fig. 7 Original and processed image to highlight expansion waves (test O221970D image 15-0931)

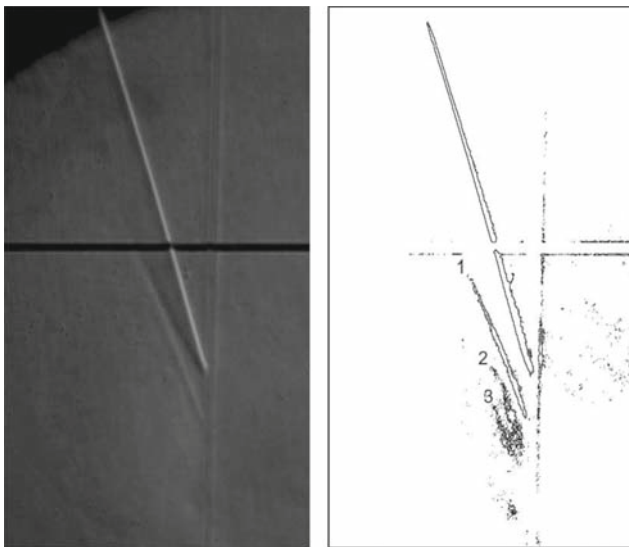


Fig. 8 Original and processed image to highlight shocklets (test L221900A image 12-0843)

the images obtained. In the first, Fig. 7, it is the darker areas which are delineated to show as white in the processed image. The lighter areas, and particularly the shocks, appear black. The supersonic patch between the first white expansion and the terminating shock are evident and are clearly followed by a second, weaker, expansion wave in front of a second dark area. Some observers have suggested that there is an indication of even a third expansion.

In the second image, Fig. 8, contours are simply drawn around all lighter areas of a given intensity value so as to highlight all the shock waves. Both the reflected shock and the first shocklet are very clear but these are followed by two further areas giving a strong indication of the existence of

further patches and their associated shocklets. Improved optical components and a shorter duration flash than those available should enable these features to be defined more clearly in future. The positions and orientations are strikingly similar to the predictions of Tesdall and Hunter [7] thereby giving a clear indication of the existence of this multiple wave system, and the presence of a more complex flow than Guderley's original four wave configuration, together with the associated multiple triple points. Thus, in terms of the discussion in Sect. 1 this could be labeled as a GMR. All computations to date have been done for an inviscid gas and the likely effects that viscosity may have on these patterns remains unclear.

4 Initial quantitative results

Since the incident wave is cylindrical, albeit appearing plane in the images due to the very large radius of curvature and the optical magnification, the triple point trajectory is thus also likely to be curved. Therefore direct quantitative measurements of the flow in the regions between the shocks become doubtful if inferred directly from the above single shock images, and the position of the pressure transducers. Fortunately, a camera taking superimposed images with adjustable time delays and a short enough exposure was available. The pixel resolution is insufficient to clearly capture the patches and shocklets but the incident, reflected, and Mach stem shocks are visible and thus the velocity and trajectory of the triple point could be established without reference to some external geometrical position. The photographs have either three or five overlaid pictures of the triple point at set time intervals between them. For an image to be considered suitable for analysis, the series of triple points must have fallen in the vicinity of two visible guides. This was initially thought to ensure a sufficiently accurate scaling factor for the velocity analysis. A typical result is given in Fig. 9

In the calculations the incident and reflected waves are assumed to be plane in the vicinity of the triple point and thus the region between them to be uniform. The radius of curvature is very large compared to the region of interest, and there is very little decay in the pressure behind the wave as recorded by the two transducers positioned as shown at A in Fig. 3 thus giving some support to this approach. A typical pressure record is given in Fig. 10.

The primary measurements taken from the photographs were the velocity of the triple point, and the wave angles between the three waves. From this data and in a frame of reference fixed in the triple point the Mach number and flow deflection angles in region 1 ahead of the incident wave; region 2, between the incident and reflected wave; and region 3, immediately behind the reflected wave, as shown in Fig. 11, could be ascertained.

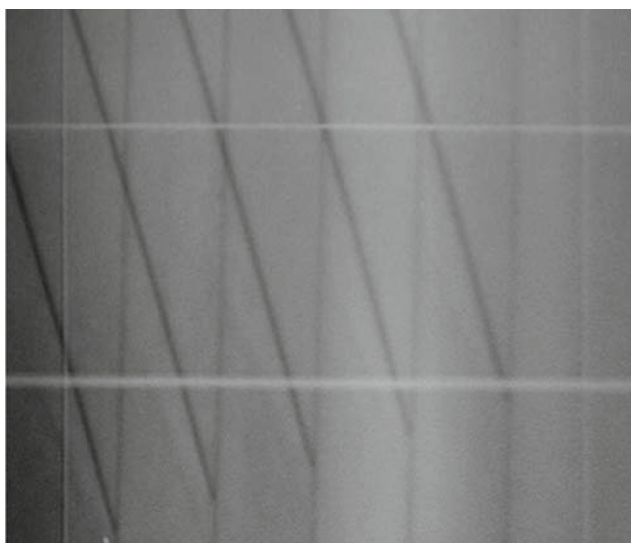


Fig. 9 Five superimposed images taken at high magnification with an interframe exposure time of 12 and 2 μ s exposure time for each image (test V221960K)

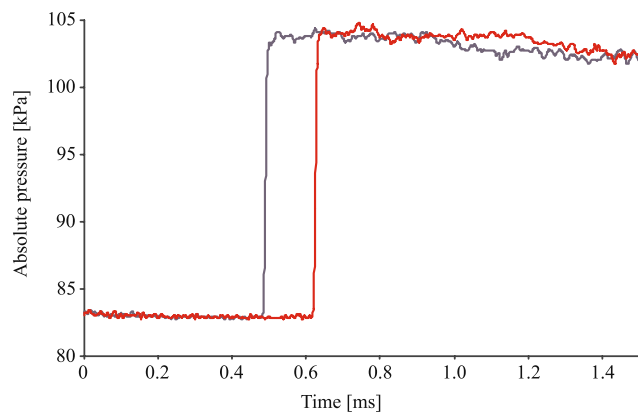


Fig. 10 Typical pressure trace record from the transducers just upstream of the window, shown at A in Fig. 3

A large number of images were analysed for the three nominal test conditions. For images containing a record of three positions of the shock, measurements were taken between the first and second record, the second and third record, and the first and third record. For the tests with five shock positions measurements were taken between the first, third, and fifth record. From these the velocity and trajectory of the triple point, as well as the relative position of the reflected shock were determined. This gives sufficient information to calculate the Mach numbers in the three regions shown in Fig. 11 as well as the respective flow directions. The results from the three determinations for each test were then averaged and are presented in Fig. 12 for each of three nominal driver pressures.

The analysis showed a significant amount of scatter in all cases, partly due to variability in diaphragm rupture, partly

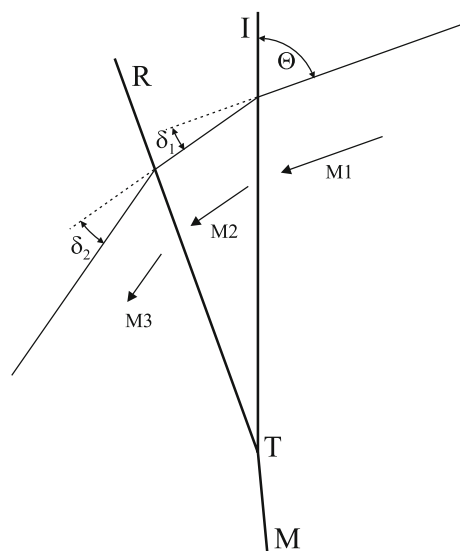


Fig. 11 Sketch of the three-shock geometry

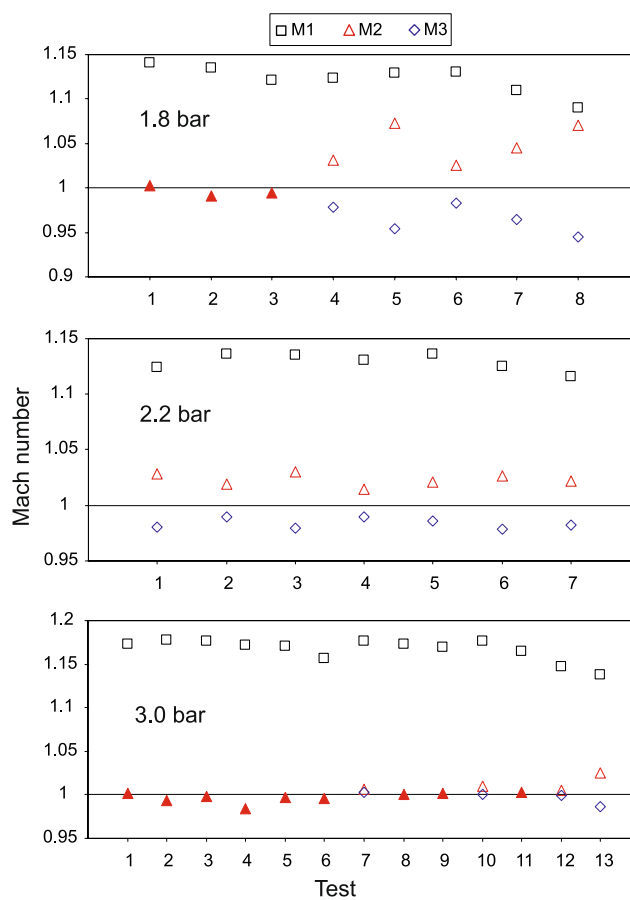


Fig. 12 Mach number data for tests at the three driver pressures

due to measurement error of the images, many of which were of lesser quality than that given in Fig. 9, and partly due to some optical distortion evidenced in some images. In all cases the reflected wave is seen to be very weak so that the

variability in the second decimal in the Mach number becomes important. This variability was of significant concern as it showed in a significant number of cases that the Mach number ahead of the reflected wave was subsonic, and in a couple of cases even if it were supersonic, that applying the shock equations with the measured angles, gave unrealistic results in that the Mach number behind the shock was higher than that in front of it. This is clearly at variance with the basic physics. The number of tests where this occurred are shown in the plots with the markers for M_2 filled and the corresponding calculated, and non-physical, values for M_3 not shown. Three of the eight tests in the upper plot failed and nine of the thirteen for the higher driver pressure.

Two reasons for these discrepancies suggest themselves. First, that of experimental error, and the effect that small variations in Mach number near sonic conditions would have in changing a flow from supersonic to subsonic. Second, and more significantly, that the slight curvature of the incident wave caused an acceleration of the flow between the incident and reflected wave to pass from subsonic to supersonic as occurs immediately behind the reflected wave. This latter issue was regarded as sufficiently important to undertake a highly detailed examination of the causes of the variability. The case with a 2.2 bar driver pressure was chosen for more detailed examination because of the consistency previously shown for this case.

5 Refined results

Whereas in the early tests the scale of the experiment was ascertained from the single distance between the horizontal threads positioned across the window and measured by a ruler, new threads were positioned in machined grooves 50 mm apart at an accuracy of better than 0.1 mm. These were then photographed and a correction matrix set up to correct the position of any point on the image to a correct physical coordinate. The camera was also sent back to the manufacturer for checking. Images were taken at larger magnification so as to increase the total distance traveled and time taken in order to improve resolution of measurement. The main calibration issue and source of error was found to be barrel distortion of the image. This effect was identified when it was noticed that the corners of the the blocks created by the new grid in the image were not square, particularly toward the periphery of the image, an example of which is given in Fig. 13. This was attributed to distortion by the objective lens of the schlieren system. In order to overcome this a two-stage spline method was used to determine the actual co-ordinates of the triple points and nearest intersection with the grid of the associated shock waves for the multi-frame images. Measurements were made on each image of the co-ordinates of the 16 grid intersection points using image processing software

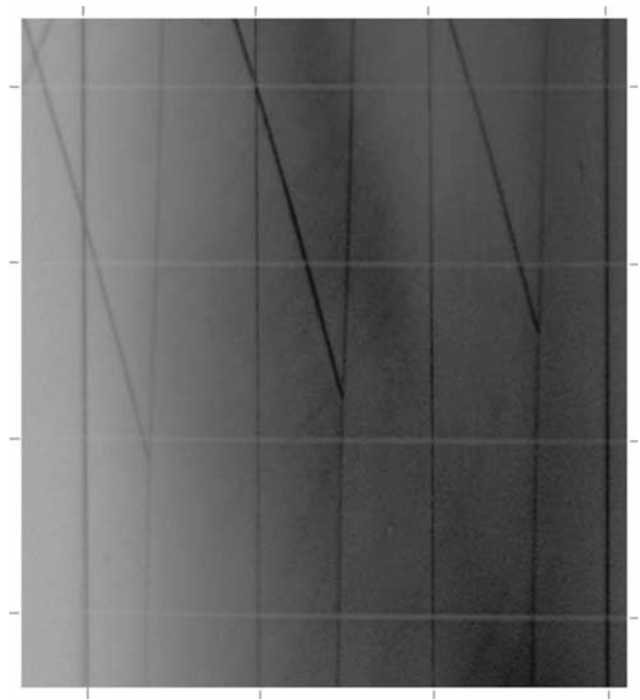


Fig. 13 Test C221620D1, Three 2 μ s exposures, 152 μ s between exposures. Grid locations shown on edges of image are at 50 mm pitch

such as Adobe Photoshop. These co-ordinates were recalculated relative to the co-ordinate of a corner of the grid pattern taken as the origin. Polynomials were then calculated for the variations of the horizontal co-ordinates versus the actual grid along each of the horizontal grid lines and of the vertical co-ordinates versus the actual grid along each of the vertical grid lines. The second stage of the spline method determined the variation of the coefficients of the horizontal co-ordinate polynomials with vertical position and vice versa for the vertical co-ordinate polynomial coefficients. This method was used instead of a hemispherical mapping function as it was not known if the distortion was truly spherical in nature.

Once these functions were derived a measured intersection point of the shock wave with the grid could be transformed into a true geometric value by using, for example, the measured value of the horizontal co-ordinate relative to the origin to determine the values of the coefficients of the vertical co-ordinate polynomial along a vertical grid line through that point and hence the true value of the vertical co-ordinate relative to the reference corner. The corrected results were then analysed in the same manner as before, with the three measurements taken off each image averaged. The results are given in Table 1.

The first point to note is that the reflected shock is very weak and the deflection through it is correspondingly small, about a quarter of a degree. The consistency between the three tests is satisfactory, and gives greater confidence in the results than the earlier uncalibrated values, and no anomalies

Table 1 Particular results, test series C221620, 2.2 bar driver pressure

Image number	Avg. M_1	Avg. M_2	Avg. M_3	Θ (deg)	δ_1 (deg)	δ_2 (deg)
C1	1.134	1.031	0.983	68.8	1.84	0.22
D1	1.129	1.030	0.980	69.0	1.72	0.22
E1	1.124	1.048	0.961	68.0	1.32	0.27

appeared using this technique. It should be emphasised, however, that the primary assumption made with these calculations is that the waves are plane. This is not strictly true for either the incident or reflected waves where the curvature in the latter case results in an acceleration to sonic conditions, and beyond. The photograph of Fig. 13 suggests that not too great an error should result from the plane shock assumption.

Since the reflected shock is so very weak, giving a small pressure change between regions 2 and 3, and the slope difference between the incident wave and Mach stem so very small, the flow immediately behind the Mach stem in the vicinity of the triple point will be similar to that behind the incident wave, i.e. about Mach 1.04. Since this is supersonic the pattern is not that of a VR. The inability to detect a slipstream is thus also not surprising.

6 Conclusion

The existence of an expansion wave at the intersection of the incident shock, reflected shock, and Mach stem, and of a supersonic patch, as originally proposed by Guderley [3] for weak shock wave reflection has been further confirmed experimentally by a more extensive range of tests than given in [2]. In addition, there is now very clear evidence of the existence of multiple supersonic patches as predicted by the computations of Tesdall and co-authors [7, 9]. The incident shock strength used for those computations are similar to the current experiments. Quantitative data of the flow Mach numbers under the assumption of plane waves has been obtained using

superimposed imaging. It is shown that the reflected wave is very weak with the flow Mach number ahead of it being less than 1.05 in a reference frame fixed in the primary triple point. This requires the use of very sensitive optical arrangements. Unless great care is taken in calibration and testing experimental variability can result in obtaining some data that is non-physical under the plane wave assumption. Detailed calibrated tests were conducted to explore whether these could be attributed to incident wave curvature. Higher resolution tests taken with a shorter duration light source would be beneficial to explore the flow patterns in more detail. In view of the small scale of the region of interest and the close juxtaposition of the many components of the multiple patches, an attempt to explore possible effects of viscosity would be desirable, since all numerical simulations to date have been for inviscid flows.

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